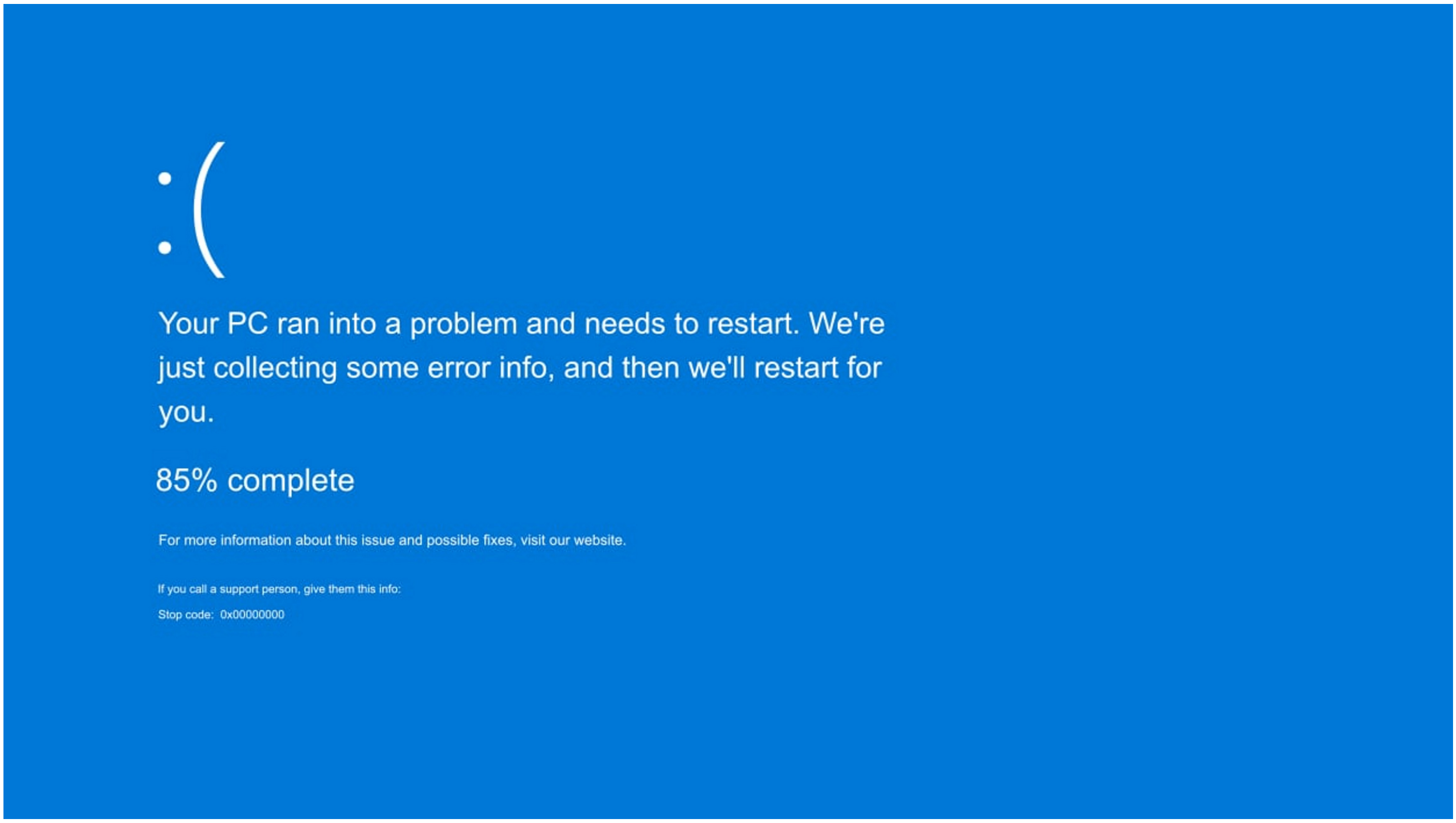
Friday, July 4, 2025 3:18 PM



💻 What is BSOD (Blue Screen of Death)?

BSOD is a **critical system crash** screen shown by **Windows** when the system encounters a **fatal error** it can’t recover from.
🧠 It’s officially called a **STOP error** or **bug check**.

🧨 Common Causes of BSOD

Cause	Example
🧩 Faulty Drivers	Incompatible graphics or network driver
⚙️ Hardware Issues	RAM, hard disk, or overheating problems
🐞 Software Bugs	Antivirus or corrupted system files
🧬 BIOS or Firmware Problems	Outdated or misconfigured BIOS
📦 Corrupt Windows System Files	Missing/damaged files (like ntoskrnl.exe)
🔒 Malware or Rootkits	Malware tampering with system kernel

📷 What Does a BSOD Look Like?

- A blue screen with an error message like:

Your PC ran into a problem and needs to restart
😢
Stop Code: DRIVER_IRQL_NOT_LESS_OR_EQUAL
🛠️ It also generates a **dump file** (minidump) and **event log** entry.

🔍 How to Find the Reason for BSOD

🔹 Step 1: Note the STOP Code

Look at the error code on the blue screen:

- Examples:
  - IRQL_NOT_LESS_OR_EQUAL
  - PAGE_FAULT_IN_NONPAGED_AREA
  - 0x0000007E

You can Google the code to identify the probable cause.

🔹 Step 2: Use Event Viewer

1. Press Win + X → Choose **Event Viewer**
2. Go to:

   Windows Logs → System

3. Filter or look for **"BugCheck"** or **critical events**.

🔹 Step 3: Analyze the Minidump File

Windows stores crash data in:

C:\Windows\Minidump\
You can open .dmp files using tools like:
✅ **BlueScreenView** (free and easy)
✅ **WinDbg** (Microsoft's official tool)
These show:

- **STOP code**
- **Driver involved**
- **Faulting module**

🔹 Step 4: Check Reliability Monitor

1. Press Win + R → type perfmon /rel
2. See a timeline of system stability.
3. Red ❌ indicates crash days — click to view details.

🔹 Step 5: Use Windows Memory Diagnostic

To check faulty RAM:

1. Press Win + R → type mdsched.exe
2. Choose "Restart now and check for problems"

🔧 Troubleshooting Tips

Problem Type	Fix
🔄 Faulty Driver	Update/reinstall using **Device Manager**
💿 Disk Errors	Run chkdsk /f /r in CMD
🔍 System File Error	Run sfc /scannow and DISM /Online /Cleanup-Image
🧠 RAM Issue	Replace/repair RAM if memory diagnostic fails
🔥 Overheating	Clean fan, reapply thermal paste
🧬 BIOS Issue	Update BIOS (from manufacturer website) carefully
🛡️ Malware	Scan using Defender or Malwarebytes

🧪 Example Scenario

You get a BSOD: Stop Code: MEMORY_MANAGEMENT

1. Check **Minidump** → shows faulty driver = ntoskrnl.exe

2. Run mdsched.exe → shows RAM error
3. Replace the faulty RAM → BSOD resolved ✅

🧾 Summary

Tool/Step	Purpose
📷 STOP Code on BSOD	Identify error type
🗂️ Minidump (.dmp)	Analyze crash details
🔎 Event Viewer	Check system logs
📈 Reliability Monitor	View crash history
🧠 Memory Diagnostic	Test for RAM issues
🧰 BlueScreenView/WinDbg	Analyze dump files